

Vendor Performance & Inventory Optimization Analysis

1. Executive Summary

This report evaluates vendor performance and inventory health to identify opportunities for profit maximization and cost reduction. The analysis revealed that a significant amount of capital (approx. \$2.71M) is currently locked in unsold inventory, driven largely by low-turnover vendors, while several high-margin brands remain under-promoted. To optimize overall profitability, the business must reallocate marketing budgets toward high-margin "Target Brands" and strategically reduce bulk order quantities from vendors contributing heavily to dead stock.

2. Methodology & Technical Stack

To ensure a robust and scalable analysis, this project was executed through an end-to-end data pipeline, transforming fragmented raw transactional data into actionable business intelligence. The workflow was divided into four key phases:

- **Data Architecture & Centralization (SQLite):** Established a localized relational database (inventory.db) to ingest and centralize highly fragmented datasets. This included mapping out relationships across multiple raw tables (Sales, Purchases, Vendor Invoices, and Purchase Prices) containing over 10,000+ transactional records.
- **Data Transformation & Aggregation (Advanced SQL):** Engineered a heavy data extraction pipeline using SQL. Leveraged **Common Table Expressions (CTEs)** and complex **LEFT JOINS** to aggregate multilevel data (e.g., merging freight costs, purchase volumes, and sales revenue) into a single, comprehensive "Source of Truth" summary table (vendor_sales_summary).

Data Cleaning & Feature Engineering (Python & Pandas): Developed a Python-based processing script to clean the aggregated database. Handled missing values, standardized categorical text features, and engineered critical new business KPIs. Calculated features included *Gross Profit*, *Profit Margin (%)*, *Stock Turnover Ratio*, and *Sales-to-Purchase Ratio* using vectorized Pandas and NumPy operations for computational efficiency.

Summary Statistics Insights:

Negative & Zero Values:

Gross Profit: Minimum value is -52,002.78, indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.

Profit Margin: Has a minimum of -00, which suggests cases where revenue is zero or even lower than costs.

Total Sales Quantity & Sales Dollars: Minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock.

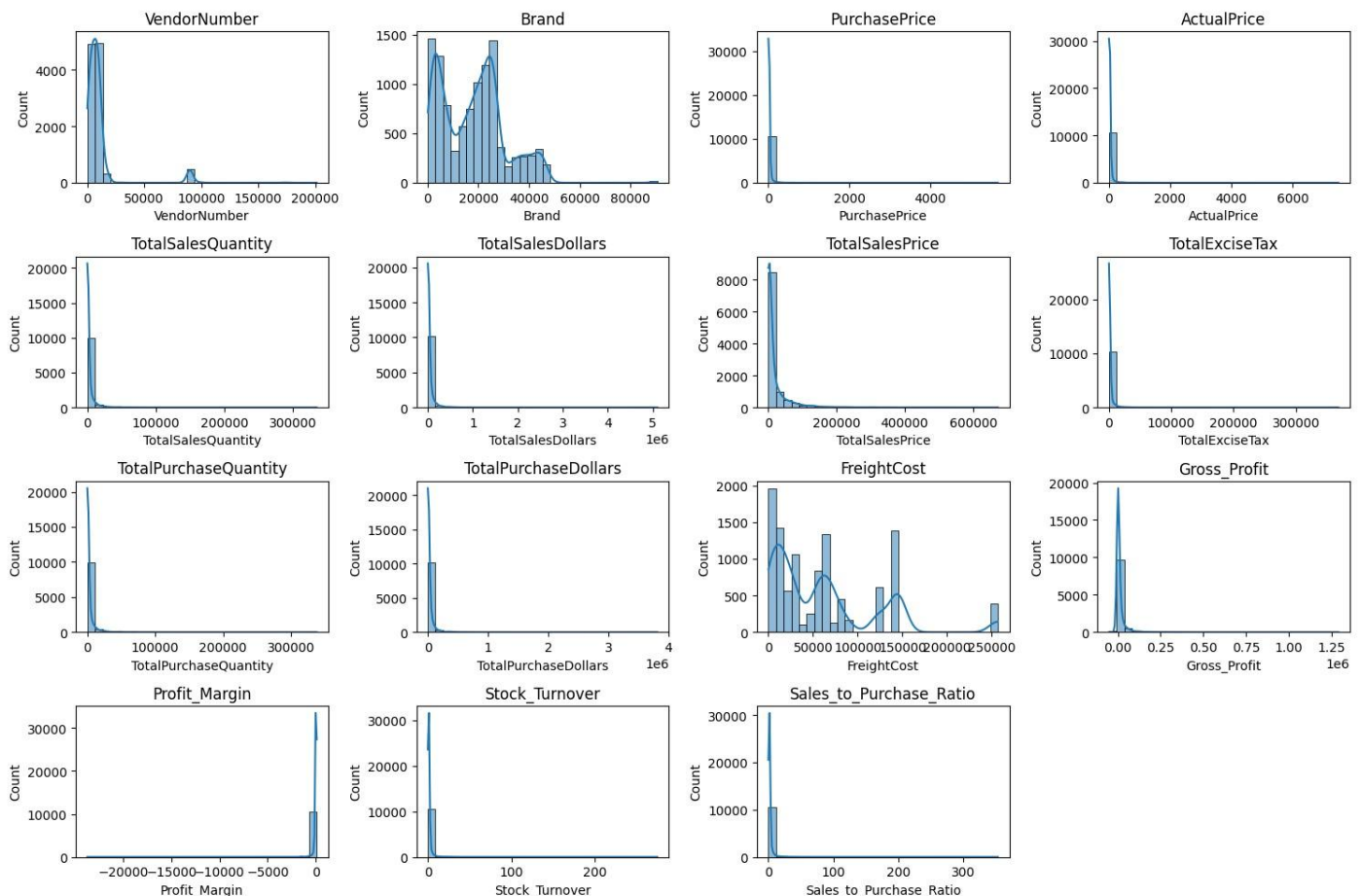
Outliers Indicated by High Standard Deviations:

Purchase & Actual Prices: The max values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating potential premium products.

Freight Cost: Huge variation, from 0.09 to 257,032.07, suggests logistics inefficiencies or bulk shipments.

Stock Turnover: Ranges from 0 to 274.5, implying some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that Sold quantity for that product is higher than purchased quantity due to either sales are being fulfilled from older stock.

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.360000	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	35.643671	148.246016	0.490000	10.990000	15.990000	28.990000	7.499990e+03
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.000000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000000	4.800000	46.570000	418.650000	3.682428e+05
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710000	453.457500	3655.465000	20738.245000	3.811252e+06
FreightCost	10692.0	61433.763214	60938.458032	0.090000	14069.870000	50293.620000	79528.990000	2.570321e+05
Gross_Profit	10692.0	12132.381048	46224.337964	-52002.780000	52.920000	1399.640000	8660.200000	1.290668e+06
Profit_Margin	10692.0	-15.620770	443.555329	-23730.638953	13.324515	30.405457	39.956135	9.971666e+01
Stock_Turnover	10692.0	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	2.745000e+02
Sales_to_Purchase_Ratio	10692.0	2.504390	8.459067	0.000000	1.153729	1.436894	1.665449	3.529286e+02



Data Filtering:

To enhance the reliability of the insights, we removed inconsistent data points where:

Gross Profit ≤ 0 (to exclude transactions leading to losses).

Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).

Total Sales Quantity = 0 (to eliminate inventory that was never sold).

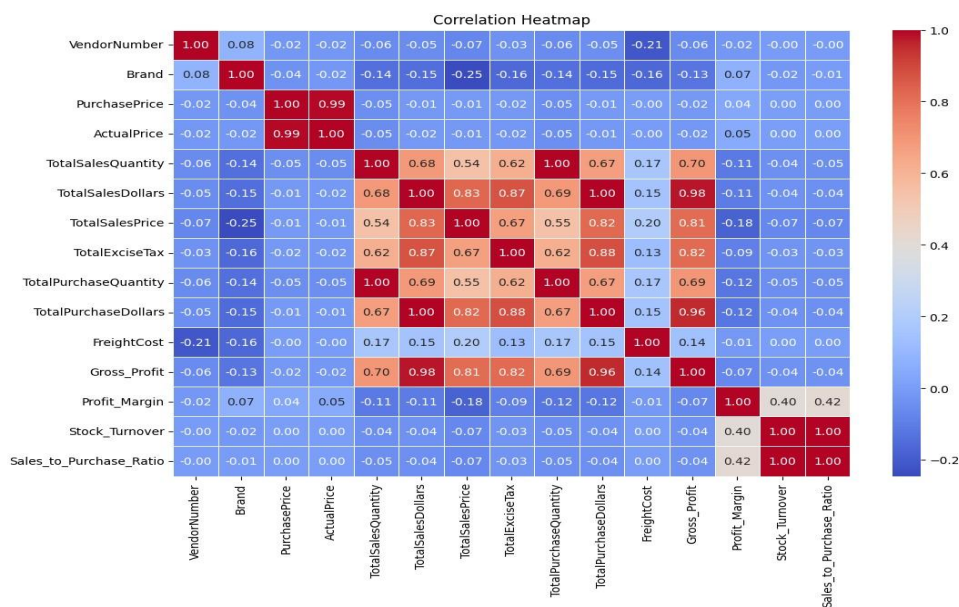
Correlation Insights:

PurchasePrice has weak correlations with TotalSalesDollars (-0.012) and GrossProfit (-0.016), suggesting that price variations do not significantly impact sales revenue or profit

Strong correlation between total purchase quantity and total sales quantity (0.999), confirming efficient inventory turnover.

Negative correlation between profit margin & total sales price (-0.179) suggests that as sales price increases, margins decrease, possibly due to competitive pricing pressures.

StockTurnover has weak negative correlations with both GrossProfit (-0.038) and ProfitMargin (-0.055), indicating that faster turnover does not necessarily result in higher profitability.



Vendors buying in bulk (Large Order Size) get the lowest unit price (\$10.78 per unit), meaning higher margins if they can manage inventory efficiently.

The price difference between Small and Large orders is substantial (~72% reduction in unit cost)

This suggests that bulk pricing strategies successfully encourage vendors to purchase in larger volumes, leading to higher overall sales despite lower per-unit revenue.

The confidence interval for low-performing vendors (40.48% to 42.62%) is significantly higher than that of top-performing vendors (30.74% to 31.61%).

This suggests that vendors with lower sales tend to maintain higher profit margins, potentially due to premium pricing or lower operational costs.

For High-Performing Vendors: If they aim to improve profitability, they could explore selective price adjustments, cost optimization, or bundling strategies.

For Low-Performing Vendors: Despite higher margins, their low sales volume might indicate a need for better marketing, competitive pricing, or improved distribution strategies.

3. Key Findings & Data Insights

A. Inventory Health & Capital Allocation

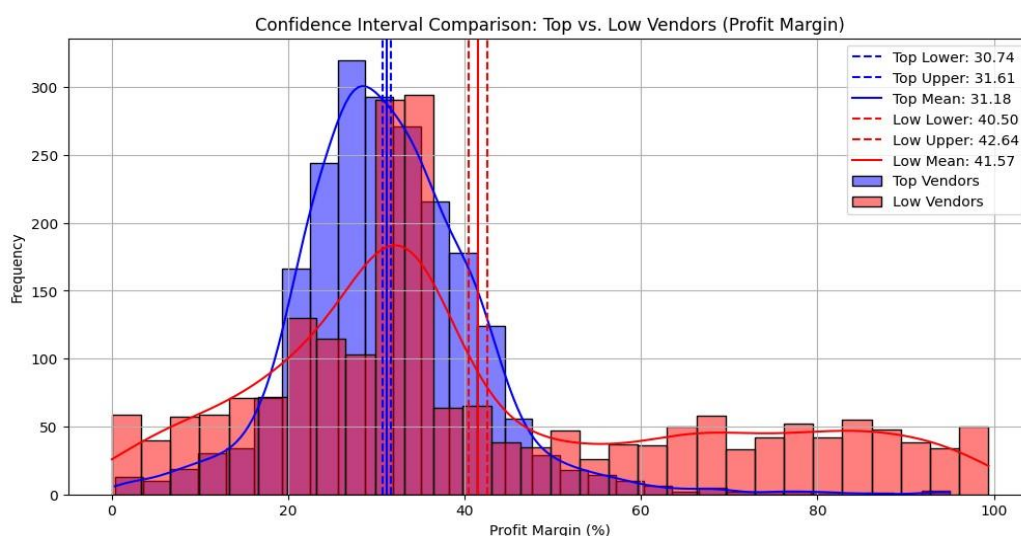
- **Locked Capital:** An estimated **\$2.71M** is tied up in unsold inventory. Minimum sales quantities for several products are 0, indicating the presence of slow-moving or completely obsolete stock.
- **Turnover Anomalies:** Stock turnover ranges drastically from 0 to 274.5. Values exceeding 1.0 indicate that current sales are being fulfilled from older, historical stock rather than recent purchases.
- **Turnover vs. Profitability:** Statistical correlation shows a weak negative relationship between Stock Turnover and Profit Margin (-0.055). This indicates that simply moving inventory faster does not automatically translate to higher profitability.

B. Pricing Dynamics & Bulk Purchasing

- **The Impact of Bulk Orders:** Vendors purchasing in bulk (Large Order Size) achieve the lowest unit price at **\$10.78 per unit**. This represents a substantial **~72% reduction** in unit cost compared to small orders.
- **Price Elasticity:** There is a negative correlation (-0.179) between Profit Margin and Total Sales Price. As the sales price increases, margins actually decrease, pointing heavily toward competitive pricing pressures in the market.
- **Purchase Price Independence:** Purchase Price shows exceptionally weak correlations with Total Sales Dollars (-0.012) and Gross Profit (-0.016), proving that base price variations do not inherently dictate final revenue or profit outcomes.

C. The "Profitability Paradox" (Vendor Performance)

Hypothesis testing revealed a statistically significant divergence in how top-performing and low-performing vendors generate profit.



- **Low-Volume Vendors Generate Higher Margins:** The 95% Confidence Interval for the profit margin of low-performing (low sales volume) vendors sits between **40.48% and 42.62%**. This suggests these vendors maintain premium pricing models or possess lower operational overhead.

Brands with Low Sales but High Profit Margins:			
	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournal 10	558.87	69.434752

198 rows x 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.

- **High-Volume Vendors Operate on Thinner Margins:** Conversely, top-performing vendors operate with a profit margin Confidence Interval of **30.74% to 31.61%**. Their success relies on high-volume turnover rather than high per-unit profitability.



Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost (\$10.7 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

OrderSize	UnitPurchasePrice
Small	39.057543
Medium	15.486414
Large	10.777625

Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: \$2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

VendorName	StockTurnover	VendorName	UnsoldInventoryValue
ALISA CARR BEVERAGES	0.615385	DIAGEO NORTH AMERICA INC	722.21K
HIGHLAND WINE MERCHANTS LLC	0.708333	JIM BEAM BRANDS COMPANY	554.67K
PARK STREET IMPORTS LLC	0.751306	PERNOD RICARD USA	470.63K
Circa Wines	0.755676	WILLIAM GRANT & SONS INC	401.96K
Dunn Wine Brokers	0.766022	E & J GALLO WINERY	228.28K
CENTEUR IMPORTS LLC	0.773953	SAZERAC CO INC	198.44K
SMOKY QUARTZ DISTILLERY LLC	0.783835	BROWN-FORMAN CORP	177.73K
TAMWORTH DISTILLING	0.797078	CONSTELLATION BRANDS INC	133.62K
THE IMPORTED GRAPE LLC	0.807569	MOET HENNESSY USA INC	126.48K

Profit Margin Comparison: High vs. Low-Performing Vendors

Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17%

Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.

Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.

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For High-Performing Vendors: If they aim to improve profitability, they could explore selective price adjustments, cost optimization, or bundling strategies.

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D. Operational Risks & Outliers

- **Loss-Making Transactions:** Minimum Gross Profit sits at a severe **-\$52,002.78**, with Profit Margins hitting zero or negative territory. This indicates products are actively being sold at a steep loss, likely due to heavy discounting dropping below the purchase cost.
- **Freight Inefficiencies:** Freight costs exhibit massive variance, ranging from a mere \$0.09 to extreme highs of ****\$257,032.07****. This points to severe logistics inefficiencies or highly disproportionate bulk shipment costs that require immediate auditing.
- **Premium Product Extremes:** Purchase and Actual Prices show maximum values (\$5,681.81 and \$7,499.99) that completely dwarf the means (\$24.39 and \$35.64).

4. Strategic Recommendations

1. **Reallocate Marketing for "Target Brands":** Redirect advertising budgets toward the low-volume, high-margin vendors identified in the 40%+ profit margin bracket. Increasing their sales volume will yield disproportionately higher gross profits.
2. **Liquidate Dead Stock:** Address the \$2.71M in locked capital by identifying the specific products with a "0" sales quantity. Implement aggressive bundling strategies or targeted discount campaigns to convert this obsolete inventory back into liquid cash.
3. **Audit Freight & Loss-Leaders:** Immediately investigate the logistics contracts causing the \$257K freight cost spikes. Concurrently, halt or strictly review the pricing algorithms allowing products to be sold at the -\$52K maximum loss threshold.
4. **Optimize Volume-Pricing Strategies:** For high-volume vendors currently operating on thinner margins (~31%), explore selective, incremental price adjustments or cost-optimization strategies to widen the margin without sacrificing their strong market share.